list.

OOPS → 'Procedural' ⇒ 'C'

Specification

```
class student
{
  Sid, Sname, m1, m2, m3
}
```

instance of a class

Sid, Sname, m1, m2, m3

Sid = 1001

Sname = 'vamsi'

m1 = 80

m2 = 84

m3 = 94.

Total = (m1 + m2 + m3)

= Sid1, Sid2, .. Sid100.

Sname1, Sname2 .. Sname100

Sid[100] ✓ 101

Student S100 = new student()

Student S2 = new student()

S2 ✓

Student S1 = new Student()

Class. Object

S1. ✓

Sid, Sname, m1, m2, m3

S1.Sid = 1001.

S1.Sname = 'vamsi'

S1.m1 = 80.

S1.m2 = 84

S1.m3 = 94

Sid, Sname, m1, m2, m3

S2.Sid = 1002

S2.Sname = 'xyz'

S2.m1 = 86

S2.m2 = 84

S2.m3 = 98.

Object name.attribute

```
class Student
{
  id, name, m1, m2, m3
}
```

attribute ← variable.

Student S1 = new Student

S1

id, name, ...

```
Class Staff.
{
  id, name, basic, hra, Tax
}.
```

Total = basic + hra - Tax

Staff s1 = new staff

id, name, basic, hra, Tax

S1.m1 + S1.m2 + S1.m3

S2.m1 + S2.m2 + S2.m3

```
int find_Total(a, b, c) - function.
{
  return a + b + c
}.
```

find_Total(S1.m1, S1.m2, S2.m3)

```
Class Student
{
  Sid, Sname, m1, m2, m3
  int find_Total( )
```

attributes ← Sid, Sname, m1, m2, m3

Methods ← int find_Total( )

Student S1 = new student()

S1.m1 = 86

S1.m2 = 84

attribute

```
Class Staff
{
  id, name, basic, hra, Tax.
```

attributes

Methods

int find_total( )
{
return m1 + m2 + m3
}
}

S1.basic

S1.m2 = 84
S1.m3 = 90
S1.find_total() } Meth

attributes
id, name, basic, hra, tax.
int find_total ()
return basic + hra - tax.
}

Staff t1 = new Staff()
t1.m1 = x.
t1.id
t1.basic = 5000
t1.hra = 3000
t1.tax = 2000
t1.find_total() = 6000

int
float
str
bool
complex.

Classes

Class int
{
attributes
Methods
}

i

i = 10 # int i = new int()

type(i).

object

Class str
{
upper
lower
capitalize
Center
index.
count.
strip
split
join
}

<Class int>

s = "Hello" # str s = new str()

s.upper() → 'HELLO'

x = [10, 20, 30, 40, 50]

class list # list x = new list()
{
count
index
insert
Methods
append
remove
pop
}

x.

x.count()
x.clear()
x.insert()

Class Complex
{
real, imag → Attribute
conjugate → Method.
}

x = 4 + 5j

Complex x = new complex()

x.real()

x.real → 4.
x.imag → 5
x.conjugate()

3+4j
3-4j

(4+5j)(4-5j)

16 + 25j² → 41 + 0j